

Exercise-induced rib stress fractures: influence of reduced bone mineral density

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Exercise-induced rib stress fractures have been reported frequently in elite rowers during the past decade. The etiology of rib stress fractures is unclear, but low bone mineral density (BMD) has been suggested to be a potential risk factor for stress fractures in weight-bearing bones. The present study investigated BMD in seven Danish national team rowers with previous rib stress fracture (RSF) and 7 controls (C) matched for gender, age, height, weight and training experience. Total body scan and specific scans of

the lumbar spine (L2–L4), femoral neck and distal radius were performed using a DEXA scanner. The RSF subjects showed significantly lower L2–L4 BMD: RSF: $1.22 \pm 0.05 \text{ g cm}^{-2}$ (mean $\pm$ SEM) (median 1.19 g cm^{-2} , range $1.02\text{--}1.37 \text{ g cm}^{-2}$) compared to C: $140 \pm 0.04 \text{ g cm}^{-2}$ (median 1.41 g cm^{-2} , range $1.27\text{--}1.57 \text{ g cm}^{-2}$) ($P = 0.028$). The present results suggest that low bone mineral density may be a potential risk factor for the development of exercise-induced rib stress fractures in elite rowers.

Rib stress fracture is a well-documented injury in both male and female elite rowers (Holden & Jackson, 1985; McKenzie, 1989; Brukner & Khan, 1996; Christiansen & Kanstrup, 1997; Palierne et al., 1997; Hickey et al., 1997; Bojanic & Desnica, 1998; Karlsson, 1998; Galilee-Belfer & Guskiewicz, 2000; Wajswelner et al., 2000). For recent review see (Warden et al., 2002).

An incidence of as high as 12% has been reported in Danish national team rowers (Christiansen & Kanstrup, 1997). A recent Australian study followed 84 female and 88 male elite rowers over a 10-year period (Hickey et al., 1997) and reported a total of 17 rib stress fractures.

The limited knowledge regarding the etiology of rib stress fractures in rowing has resulted in a lack of significant prophylactic interventions. Several mechanisms of injury and potential risk factors have been suggested during the past 20 years as summed up by Warden et al. (2002) who concluded that the etiology is multifactorial.

A few studies on rib stress fractures in rowers have mentioned that low bone mineral density (BMD)

may be a potential etiological risk factor (Holden & Jackson, 1985; Hickey et al., 1997), although there is no experimental evidence. In contrast, several studies have indicated that low BMD in athletes and military recruits may be a risk factor for stress fractures in the lower extremities (Pouilles et al., 1989; Myburgh et al., 1990; Bennell et al., 1996; Lauder et al., 2000). Moreover, low BMD has been demonstrated to be a predictor of non-stress rib fractures in perimenopausal women (Kroger et al., 1995). Thus, the aim of the present study was to investigate BMD in rowers with and without previous rib stress fracture to lend support to the hypothesis that rowers with previous rib stress fracture are also characterized by a reduced BMD.

Materials and methods

All athletes from the Danish national rowing team who accepted to participate in the study ($n = 29$) were weighed, had their body length measured and they filled in a questionnaire with information on gender, age, number of years of elite training and rib stress fracture history. By interviewing

the subjects who reported previous rib stress fractures it was ascertained that all previous rib stress fractures included in this study were clinically diagnosed by the national team physician and in most cases verified by a ^{99m}Tc Technetium-MDP bone scan. Two subjects were excluded from the study due to uncertain rib stress fracture status. The remaining seven subjects with a history of rib stress fractures (RSF) participated in the present study, and seven controls were carefully selected from the remaining national team rowers to match the stress fracture subjects. The subjects were matched for gender, age, height, weight and number of years of elite training. Each group consisted of five lightweight male rowers, one lightweight and one heavyweight female rower. Seventeen cases of rib stress fracture were reported in the seven rowers over the period 1994–2000.

Physical characteristics of the subjects are presented in Table 1a and b.

All subjects gave their written informed consent to participate in the experiments, which were approved by the local ethics committee. Data were collected prior to the 2000 Olympic Games in Sydney.

Experimental procedures

Measurements of BMD of total body, lumbar spine (L2–L4), femoral neck (bilaterally) and distal radius (dominant side) were performed by use of dual energy X-ray absorptiometry (DEXA) (Lunar Corporation, Madison, WI, USA). Scan type DPX-IQ, software version 4.6.6. was used for calculation of total body (fast mode), lumbar spine and femoral neck scans (high resolution, fast mode). Scan type Expert-XL, software version 1.8 was used to process the distal radius scans.

BMD results are presented in g cm^{-2} and in percent of a young adult reference population given by the DEXA software. Lean body mass (LBM) and percent body fat (BF) were obtained from the whole body DEXA scan.

Statistical analysis

All data are presented as mean $\pm$ SEM, median and range. Statistical comparisons between the RSF group and the control group were performed using the Wilcoxon signed ranks test for paired samples. All correlation analyses were performed using the Spearman's ρ test (Altman, 1991). The level of significance was set at $P = 0.05$.

Results

Although not intentionally matched with respect to body composition, the two groups showed similar LBM and BF (Table 1a and b). Because of this and the matched pairs design used in the present study, the BMD results were not adjusted for body weight or any other variables.

Total and regional BMD of the RSF subjects and the controls in g cm^{-2} are presented in Table 2a and b. Compared to the matched control group, the RSF group showed $12.9 \pm 3.8\%$ (median 12.7%, range -27.7 to $+5.6\%$) lower BMD ($P = 0.028$) in the lumbar spine (L2–L4). The corresponding lumbar

Table 1. Physical characteristics of (a) RSF subjects, (b) control subjects

Subject	Gender	Age (years)	Height (cm)	Weight (kg)	Years of training	Lean body mass (kg)	Body fat (%)	Rib stress fractures (n)
<i>(a) RSF subjects</i>								
1	Male	29	188	77	8	65.7	11.2	3
2	Male	26	190	75	8	66.0	8.8	3
3	Male	23	185	77	8	67.3	8.3	3
4	Male	20	173	77	5	64.2	13.4	2
5	Male	32	179	74	12	67.3	7.6	1
6	Female	25	171	61	3	44.3	22.4	1
7	Female	24	179	71	5	53.0	21.7	4
Mean		25.6	180.7	73.1	7	61.1	13.3	2.4
SEM		1.5	2.8	2.2	1.1	3.4	2.4	0.4
Median		25	179	75	8	65.7	11.2	3
Range		20–32	171–190	61–77	3–12	44.3–67.3	7.6–22.4	1–4
Subject	Gender	Age (years)	Height (cm)	Weight (kg)	Years of training	Lean body mass (kg)	Body fat (%)	
<i>(b) Control subjects</i>								
1	Male	28	184	78	12	70.4		5.4
2	Male	25	187	73	8	64.8		7.1
3	Male	24	183	80	8	70.8		8.8
4	Male	21	185	76	5	61.1		14.3
5	Male	26	182	74	12	65.5		7.2
6	Female	27	167	61	1	46.8		19.4
7	Female	27	175	72	4	55.0		18.5
Mean		25.4	180.4	73.4	7.1	62.1		11.5
SEM		0.9	2.7	2.3	1.6	3.2		2.2
Median		26	183	74	8	64.8		8.8
Range		21–28	167–187	61–80	1–12	46.8–70.8		5.4–19.4

RSF subject number 1 (a) and control subject number 1 (b) are matched etc.

Table 2. Total and regional BMD (g cm^{-2}) in (a) RSF subjects, (b) control subjects

Subject	Total Body	L2-L4	Femoral neck average	Distal radius
<i>(a) RSF subjects</i>				
1	1.17	1.19	1.03	0.31
2	1.16	1.18	0.95	0.37
3	1.31	1.35	1.12	0.45
4	1.23	1.13	1.07	0.47
5	1.29	1.37	1.09	0.48
6	1.19	1.29	1.05	0.34
7	1.10	1.02	0.88	0.25
Mean $\pm$ SEM	1.21 ± 0.03	1.22 ± 0.05^a	1.03 ± 0.03	0.38 ± 0.03
Median	1.19	1.19	1.05	0.37
Range	1.10–1.31	1.02–1.37	0.88–1.12	0.25–0.48
<i>(b) Control subjects</i>				
1	1.32	1.41	1.11	0.48
2	1.17	1.30	0.84	0.39
3	1.24	1.27	1.21	0.47
4	1.20	1.39	0.96	0.40
5	1.34	1.57	1.45	0.47
6	1.30	1.47	1.20	0.35
7	1.34	1.41	1.11	0.48
Mean $\pm$ SEM	1.27 ± 0.03	1.40 ± 0.04	1.13 ± 0.07	0.43 ± 0.02
Median	1.30	1.41	1.11	0.47
Range	1.17–1.34	1.27–1.57	0.84–1.45	0.35–0.48

RSF subject number 1 (a) and control subject number 1 (b) are matched etc.
^aSignificant difference from control subjects ($P = 0.028$).

spine BMD values expressed as a percent of a young adult reference population was $99.1 \pm 3.8\%$ (median 96%, range 85–111%) and $115.3 \pm 3.7\%$ (median 114%, range 108–127%) for the RSF and the controls, respectively. Also the scans of the total body, the femoral neck and the distal radius showed mean and median values close to normal in the RSF subjects: $101.1 \pm 1.9\%$ (median 101%, range 96–107%), $98.2 \pm 2.8\%$ (median 100%, range 88.5–107.5%) and $92.1 \pm 7.2\%$ (median 90%, range 66–114%), respectively, whereas the corresponding mean and median values for the controls were consistently above normal: $106.9 \pm 3.2\%$ (median 110%, range 96–119%), $108.2 \pm 7.3\%$ (median 112.5%, range 78.5–136%) and $105.4 \pm 4.9\%$ (median 111%, range 91–126%).

No significant relationships between neither body fat nor lean body mass and BMD were observed in the 10 male subjects.

Discussion

Consistently lower BMD was observed in the RSF subjects at all scan locations, which reached statistical significance in the lumbar spine. Therefore a possible relationship between reduced BMD and an elevated incidence of rib stress fractures can be suggested. However, the small subject sample and the retrospective nature of this study including the risk of bias regarding the subjects' ability to recall

the exact number of rib stress fractures and how the diagnosis was made must be remembered. Therefore additional evidence for low BMD as an etiological risk factor for the development of exercise-induced rib stress fractures in elite rowers is yet to be provided.

To our knowledge no previous study has examined the possible relationship between BMD and the incidence of rib stress fractures in elite rowers. The recent review by Warden et al. (2002) on the etiology of rib stress fractures in rowers found only a single abstract, which was a preliminary presentation of the present data (Vinther et al., 2002). Several studies have reported a correlation between BMD and the incidence of stress fractures of the lower extremities in athletes and military recruits (Pouilles et al., 1989; Myburgh et al., 1990; Bennell et al., 1996; Crossley et al., 1999; Beck et al., 2000; Lauder et al., 2000). Two studies investigated men and women separately, one including track and field athletes (Bennell et al., 1996) and one including military recruits (Beck et al., 2000). Both Bennell et al., (1996) and Beck et al., (2000) found significantly lower BMD in stress fracture cases compared to controls in women ($n = 10$ stress fracture cases/36 controls and 37/626 respectively) but not in men ($n = 10/39$ and 38/587, respectively). A study including active-duty army women (Lauder et al., 2000) and one including athletes of both genders (72% women) (Myburgh et al., 1990) also demonstrated a significant correlation between reduced BMD and the stress fracture incidence. One study on male subjects found a significant correlation between low femoral BMD and the incidence of calcaneal and femoral stress fractures (Pouilles et al., 1989). In contrast, another study failed to see any correlation between tibial BMD and tibial stress fractures in male runners (Crossley et al., 1999). The latter study found that bone geometry was significantly correlated to fracture incidence, which was also the case for both men and women recruits in the study conducted by Beck et al. (2000). The ability of bone to resist repetitive stress forces is dependent on bone geometry as well as bone material properties (Crossley et al., 1999; Beck et al., 2000). Thus, measurements of rib geometry and specific BMD assessments of the ribs in elite rowers should be included in future investigations to better elucidate this potential risk factor. It was not possible to perform specific BMD measurements of the ribs since this particular part of the whole-body scan was confounded by the scapula and the clavicle. Nevertheless, the results indicate a substantial difference in bone material properties between RSF subjects and controls.

An interesting finding was that the RSF subjects showed BMD values that were compatible to the young adult reference population. In contrast, a

majority of the control subjects showed BMD values that exceeded that of the young adult reference population. Since the rowers were individually matched also with respect to number of years of training, this finding could imply that in addition to genetic differences some subjects might experience an adaptive increase in BMD with training, while others do not. This notion is supported by the results of Pouilles et al. (1989), who observed increased femoral BMD during 12 weeks of recruit training in 16 subjects while 12 subjects remained unchanged and 7 experienced bone loss.

Relationships between BMD and BF, and LBM for female subjects have been debated (Helge & Kanstrup, 2002). However, the RSF subjects did not differ from controls with respect to BF or LBM. Since both BF and LBM appears to be gender dependent, BMD-BF and BMD-LBM correlation analysis only was performed for the ten male subjects in the present study. No significant correlations were observed.

In the present material five out of seven RSF subjects were male lightweight rowers. This group of rowers frequently use fasting and restriction of food intake to reduce bodyweight and remain within certain weight limits. Talbott & Shapses (1998) found that a 24-h fast resulted in a 20–25% reduction in selected markers of bone turnover (serum osteocalcin and urinary pyridium cross-links) in male lightweight rowers indicating a suppression in both bone formation and bone resorption. Consequently, it is possible that weight control methods employed by lightweight rowers can result in periods of lowered bone turnover, which in turn could affect the ability of bone to adapt to training. In addition to nutritional factors, hormonal factors have been suggested to influence BMD in male endurance athletes. Lowered levels of testosterone have been reported in endurance athletes compared to sedentary controls (Wheeler et al., 1984; Hackney et al., 1998). Further, it has been speculated that low testosterone could lead to demineralization and reduced BMD in male endurance athletes (Bennell et al., 1996; Hackney, 2001). Consequently, both nutritional and hormonal factors need to be investigated to further elucidate if these

factors influence BMD and hence constitute a potential risk factor for rib stress fracture development in this population.

Perspectives

The present comparison of BMD to a young adult reference population may constitute a valuable outcome measure, as it allows for an individual evaluation of BMD in single subjects, i.e. in individual rowing athletes from the national team. Importantly, this enables the medical team and trainers to identify rowers with low BMD. If BMD definitively proves to be an etiological risk factor for the development of rib stress fractures in elite rowers, the intensity and the frequency of the training as well as restricted food intake and fasting could be potential risk factors in subjects with low BMD. It is possible that rowers with different bone mineral characteristics will respond differently to a given magnitude and intensity of training, hence resulting in the development of rib stress fractures in individuals with low BMD.

Conclusion

Elite rowers with exercise-induced rib stress fractures showed significantly reduced BMD of the lumbar spine (L2–L4) compared to matched controls. This finding suggests that low BMD may be a risk factor for the development of rib stress fractures in rowing. However, prospective studies including larger materials are needed to more conclusively determine if BMD and/or bone geometry are potential risk factors for the development of exercise-induced rib stress fractures in elite rowers.

Key words: body fat, lean body mass, incidence.

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